ROUTE B FOR THE MIDPOINT RVM SYSTEM: AN A PRIORI BOUND FOR QUASI-SOLUTIONS, AND A KADEC–AVDONIN CRITERION WITH BLOCKS OF VARIABLE LENGTH

ROTKIV AYNÄSAC

Abstract. We work with real lists $\boldsymbol{\kappa} = (\kappa_k)_{k\ge 1}$ at bounded distance from the midpoint Riemann–von Mangoldt clock (τ_k) and with the rung system that couples such a list to atomic masses on $\{\log n\}$. Three groups of results are proved. (1) Route B: near-solutions of finite sections, taken in a fixed box, have a subsequence converging to an exact solution; under a uniqueness hypothesis in the box, the whole sequence converges, and the limit is the zeta list. Conversely, if condition (D) of the main manuscript holds, the section residuals tend to zero. If the masses are prescribed to be $\Lambda(n)/(2\sqrt{n})$, the uniqueness hypothesis is not needed: under a convention on the Laplace transform of the residual, any exact solution at bounded distance from the clock is the zeta list and forces the Riemann hypothesis, so this route reduces to a single existence statement, which we also formulate as a staged extension lemma. (2) For quasi-solutions with Laplace residual of size $\varepsilon \log T$, the counting discrepancy with respect to the clock is $O((1+\varepsilon)\log T)$ with absolute constants. As a consequence, the displacement of every sufficiently high ordinate is at most $2\pi(a_1 + a_2\varepsilon) + o(1)$. (3) A sufficient condition of Kadec–Avdonin type for the Muckenhoupt (A_2) property of a generating function. The averaging blocks may have arbitrary, varying length, at the price of a bounded-conjugate condition on the oscillation inside the blocks. The classical $\frac{1}{4}$ theorems of Kadec and Avdonin are recovered with the sharp constant, and a variable-length version of Avdonin's theorem follows. For lists near the clock the averaging condition holds automatically with blocks of $\asymp \log^2 T$ points. For the zeta list the remaining condition fails: unconditionally, no generating function of the unfolded zeta ordinates satisfies (A_2) on any horizontal line.

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1. Introduction

The main manuscript constructs a synthetic zeta system on the midpoint clock τ_k, defined by $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, and reduces the remaining step to condition (D). Route B approaches (D) without constructing the zeta ordinates. One solves finite sections of the rung system (2.4), which imposes simultaneously a real critical spectrum and atomic arithmetic on $\{\log n\}$. If the residuals of these sections tend to zero in a fixed box, compactness yields an exact solution. The uniqueness theorem `thm:free-masses` then identifies that solution as the list of zeta ordinates with the von Mangoldt masses. In this scheme the ordinates γ_k are the output of an inverse problem, not an input. Section 3 makes this precise and isolates the two statements that remain open: existence of sections with vanishing residual, and uniqueness in the box without the third condition of $\mathscr{O}$. Section 3.1 shows that prescribing the von Mangoldt masses removes the second: uniqueness then follows from Hadamard's theorem and the functional equation. Section 3.2 formulates the existence part as an extension lemma for staged sections started at the clock.

Section 4 proves an a priori bound for quasi-solutions. Section 5 proves a Kadec–Avdonin criterion with blocks of variable length, which is the tool needed for a Fredholm/stability formulation of the linearised problem. Section 6 applies that criterion to lists near the clock and proves that, for the zeta list, the uniform version fails.

References of the form `thm:free-masses` point to labels of the main manuscript `rvm_zeta_pi_mertens_v18`. Except in Section 6.3, where the zeta list is the object of study, nothing below uses the ordinates of ζ.

2. Setting

Throughout, $q(s) := s(s-1)$ and $G(s) := s(s-1)\pi^{-s/2}\Gamma(s/2)$. The smooth clock count is

$$\mathcal{N}_{\mathrm{RVM}}(t) := \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}, \qquad \mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} \qquad (t > 0). \tag{2.1}$$

It is strictly increasing on $[2\pi, \infty)$, with $\mathcal{N}_{\mathrm{RVM}}(2\pi) = -\frac{1}{8}$, and the clock points are the solutions $\tau_k > 2\pi$ of $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, $k \geq 1$ ($\tau_1 = 14.5213\ldots$). We write $N_\tau(t) := \#\{k \geq 1 : \tau_k \leq t\}$. Since $\mathcal{N}_{\mathrm{RVM}}$ is increasing,

$$N_\tau(t) = \Big\lfloor \mathcal{N}_{\mathrm{RVM}}(t) + \tfrac{1}{2} \Big\rfloor, \qquad |N_\tau(t) - \mathcal{N}_{\mathrm{RVM}}(t)| \leq \tfrac{1}{2} \qquad (t \geq 2\pi). \tag{2.2}$$

Definition 2.1 (Admissible lists)**.** For $C > 0$, a list $\boldsymbol{\kappa} = (\kappa_k)_{k\geq 1}$ is *C-admissible* if it is real and nondecreasing, $\kappa_1 \geq 2\pi$, and $|\kappa_k - \tau_k| \leq C$ for all k. Put $N_{\boldsymbol{\kappa}}(t) := \#\{k : \kappa_k \leq t\}$ (with multiplicity) and

$$P_{\boldsymbol{\kappa}}(s) := \prod_{k\geq 1}\Big(1 + \frac{q(s)}{\frac{1}{4} + \kappa_k^2}\Big), \qquad \zeta_{\boldsymbol{\kappa}} := \frac{P_{\boldsymbol{\kappa}}}{G}. \tag{2.3}$$

The clock list $\boldsymbol{\tau} = (\tau_k)$ gives $\zeta_{\boldsymbol{\tau}} = \zeta_{\mathrm{RVM}}$.

$P_{\boldsymbol{\kappa}}$ is entire and real on $\mathbb{R}$, and $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$. Its zeros are exactly $\frac{1}{2} \pm i\kappa_k$, with multiplicity, and it is positive on $[-1, 2]$, because $q([-1,2]) = [-\frac{1}{4}, 2]$ and $\kappa_k^2 \geq 4\pi^2$. Writing $s = \frac{1}{2} + w$, one has $\frac{1}{4} + \kappa^2 + q(s) = \kappa^2 + w^2$.

The *rung system* `eq:rung-system` for a list $\boldsymbol{\kappa}$ and real masses $\mathsf{a} = (\mathsf{a}_n)_{n\geq 2}$ is

$$\sum_{k\geq 1}\cos(\kappa_k u) + \sum_{n\geq 2}\mathsf{a}_n\,\delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{2.4}$$

with the fixed distribution k of `eq:transport-defs`. Its residual is the distribution $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a})$ defined, for $\varphi \in C_c^\infty(0,\infty)$, by

$$\langle \mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle := \sum_{k\geq 1} F_\varphi(\kappa_k) + \sum_{n\geq 2}\mathsf{a}_n\varphi(\log n) - \langle \mathsf{k}, \varphi\rangle, \qquad F_\varphi(t) := \int_0^\infty \varphi(u)\cos(tu)\,du. \tag{2.5}$$

Two integrations by parts give $|F_\varphi(t)| \le \|\varphi''\|_{L^1}\, t^{-2}$. Since $\kappa_k \ge \tau_k - C$ and $\tau_k \gg k/\log(k+1)$, the series in (2.5) converges absolutely. The mass sum is finite, because $\operatorname{supp}\varphi \subset [u_1, U]$ with $0 < u_1 < U$ leaves only $n \le e^U$.

3. Route B: compactness, selection and the canonical box

For $C, M > 0$ let $\mathsf{K}(C, M)$ be the set of pairs $(\boldsymbol{\kappa}, \mathsf{a})$ with $\boldsymbol{\kappa}$ C-admissible and $|\mathsf{a}_n| \le M$ for all n. A *finite section* is an element of $\mathsf{K}(C, M)$ with $\kappa_k = \tau_k$ for $k > K$ and $\mathsf{a}_n = 0$ for $n > N$, for some K, N.

Fix finite sets $\Phi_1 \subset \Phi_2 \subset \cdots \subset C_c^\infty(0,\infty)$ whose union $\mathscr{D}$ is dense in $C_c^\infty(0,\infty)$. Such sets exist because $C_c^\infty(0,\infty)$ is separable: it is a countable strict inductive limit of separable Fréchet spaces. For $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathsf{K}(C, M)$ put

$$\rho_m(\boldsymbol{\kappa}, \mathsf{a}) := \max_{\varphi \in \Phi_m} \big|\langle \mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle\big|. \tag{3.1}$$

Definition 3.1 (Statement B)**.** *Statement $B(C, M)$* says that there are finite sections $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathsf{K}(C, M)$ with $\rho_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$.

Lemma 3.2 (Canonical box of the exact solution)**.** *Let $M_* := \max_{n\ge 2} \Lambda(n)/(2\sqrt{n})$. Then $M_* = \log 7/(2\sqrt{7}) = 0.367742\ldots < 1/e$. If condition (D) holds, the pair $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$, with $\mathsf{a}^\zeta_n := \Lambda(n)/(2\sqrt{n})$, solves* (2.4) *and lies in $\mathsf{K}(C_\eta, M_*)$, where C_η is the constant of* `lem: bounded-displacement`*.*

Proof. $x \mapsto \log x/(2\sqrt{x})$ increases on $[1, e^2]$ and decreases on $[e^2, \infty)$, with $e^2 = 7.389\ldots$. Hence among primes its maximum is at $p = 7$ or $p = 11$; the values are $0.367742\ldots$ and $0.361540\ldots$. A prime power p^j with $j \ge 2$ gives $\log p/(2p^{j/2}) < \log p/(2\sqrt{p}) \le M_*$. The bound by $1/e$ is the continuous maximum, attained at $x = e^2$. Under (D), $\zeta_{Z1} = \zeta$, so all zeros of ζ are $\frac12 \pm i\gamma_k$. The rung system with list $\boldsymbol{\gamma}$ and masses a^ζ is then the distributional explicit formula `thm:distributional-EF`. Finally, $|\gamma_k - \tau_k| \le C_\eta$ by `lem:bounded-displacement`, and $\gamma_1 = 14.13\ldots \ge 2\pi$. □

Corollary 3.3 ((D) implies B)**.** *If (D) holds, then $B(C, M)$ holds for every $C \ge C_\eta$ and $M \ge M_*$.*

Proof. Let $(\boldsymbol{\kappa}^{(K,N)}, \mathsf{a}^{(K,N)})$ be the truncation of $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$. Its list is the nondecreasing rearrangement of the multiset $\{\gamma_1, \ldots, \gamma_K\} \cup \{\tau_{K+1}, \tau_{K+2}, \ldots\}$, and its masses are a^ζ_n for $n \le N$ and 0 beyond. If an unsorted list satisfies $|\kappa_k - \tau_k| \le C$ with (τ_k) increasing, so does its nondecreasing rearrangement: the j-th smallest element is at most $\tau_j + C$ and at least $\tau_j - C$. Beyond some index the rearranged list coincides with (τ_k), and $\mathcal{R}$ depends only on the multiset. So this is a finite section in $\mathsf{K}(C_\eta, M_*)$. For fixed φ, $\langle \mathcal{R}(\boldsymbol{\kappa}^{(K,N)}, \mathsf{a}^{(K,N)}), \varphi\rangle - \langle \mathcal{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta), \varphi\rangle = \sum_{k>K}(F_\varphi(\tau_k) - F_\varphi(\gamma_k))$ as soon as $e^U \le N$. This tends to 0 as $K \to \infty$. Since Φ_m is finite and $\mathcal{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ by Lemma 3.2, K_m, N_m can be chosen with $\rho_m \le 1/m$. □

Proposition 3.4 (Compactness and passage to the limit)**.** *Let $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathsf{K}(C, M)$ with $\rho_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$. Then a subsequence converges coordinatewise to some $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathsf{K}(C, M)$, and every such limit satisfies $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ in $\mathcal{D}'(0,\infty)$.*

Proof. Each $\kappa_k^{(m)}$ lies in the compact interval $[\tau_k - C, \tau_k + C]$ and each $\mathsf{a}_n^{(m)}$ in $[-M, M]$. A diagonal extraction over the countably many coordinates gives a coordinatewise convergent subsequence. The limit is nondecreasing, satisfies $\kappa_1 \ge 2\pi$, $|\kappa_k - \tau_k| \le C$ and $|\mathsf{a}_n| \le M$, and therefore lies in $\mathsf{K}(C, M)$.

Continuity of each pairing. Fix φ with support in $[u_1, U]$. The mass part of (2.5) involves only $n \le e^U$ and converges. For the spectral part, $|F_\varphi(\kappa_k^{(m)})| \le \|\varphi''\|_1(\tau_k - C)^{-2}$ when $\tau_k > 2C$, and $|F_\varphi| \le \|\varphi\|_1$ for the finitely many other k. The majorant is summable, so dominated convergence for series gives $\sum_k F_\varphi(\kappa_k^{(m)}) \to \sum_k F_\varphi(\kappa_k)$. Hence $\langle \mathcal{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle \to \langle \mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle$ along the subsequence.

Vanishing. If $\varphi \in \mathscr{D}$, then $\varphi \in \Phi_m$ for all large m, so $|\langle \mathcal{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle| \le \rho_m \to 0$ and $\langle \mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle = 0$. The limit $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a})$ is a single distribution, hence continuous on $C_c^\infty(0,\infty)$. It vanishes on the dense set $\mathscr{D}$, so it vanishes identically. No uniform bound on the residuals is needed for this step. □

Hypothesis 3.5 (Uniqueness in the box)**.** $U(C, M)$: the only $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathsf{K}(C, M)$ with $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ is $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$.

Remark 3.6 (What is proved and what is open in U). `thm:free-masses` proves uniqueness for lists in $\mathscr{O}$. Besides the tube and the ordering, $\mathscr{O}$ (`eq:list-space`) carries a third condition, $|\int (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}})| \le C \log X$, and a coordinatewise limit of sections does not inherit it. Hypothesis 3.5 is therefore the statement that `thm:free-masses` holds for exact solutions in $\mathsf{K}(C, M)$ without the third condition. That condition is used to bound $\zeta_{\boldsymbol{\kappa}}$ on half-planes. For an exact solution, Lemma 4.3 below with $\widehat{\mathsf{R}} \equiv 0$ gives that bound directly on $\Re s \ge 2$. Whether this suffices for the rest of the proof of `thm:free-masses` has to be checked line by line against that proof. This is not done here, and Hypothesis 3.5 is not claimed.

Corollary 3.7 (Selection of the zeta list)**.** *Assume $B(C, M)$ and $U(C, M)$, and let $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)})$ be any sequence in $\mathsf{K}(C, M)$ with $\rho_m \to 0$. Then the whole sequence converges coordinatewise to $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$. In particular $\mathcal{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$, that is, (D) holds.*

Proof. By Proposition 3.4, every subsequence has a further subsequence converging to a zero of $\mathcal{R}$ in $\mathsf{K}(C, M)$. By $\mathrm{U}(C, M)$ that zero is $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$, so the whole sequence converges to it. Finally we show that $\mathcal{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ implies (D). We use the list form of `thm:distributional-EF` in this form: $\widehat{\mathsf{R}}$ of (4.1) is the Laplace transform of a distribution of finite order on $[0,\infty)$ whose restriction to $(0,\infty)$ is $\mathcal{R}$. If $\mathcal{R} = 0$, that distribution is supported at $\{0\}$, so $\widehat{\mathsf{R}}$ is a polynomial $p(s)$. Since $\sum_n \mathsf{a}_n^\zeta n^{\frac12 - s} = -\frac12 \zeta'/\zeta(s)$, this reads $P_\gamma'/P_\gamma - (G\zeta)'/(G\zeta) = 2p$, so $P_\gamma = c\, G\zeta\, e^Q$ with $Q' = 2p$. Here $G\zeta = 2\xi$ and P_γ are entire of order 1, so $e^Q = P_\gamma/(2c\,\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \le 1$. Both P_γ and ξ are invariant under $s \mapsto 1 - s$, so $e^{Q(1-s)} = e^{Q(s)}$, which forces Q to be constant. Hence $\zeta_{\boldsymbol{\gamma}} = c'\zeta$. Both functions have residue 1 at $s = 1$ ($P_\gamma(1) = 1$ and $G(s) \sim s - 1$), so $c' = 1$. This is (D), because $\zeta_{\boldsymbol{\gamma}} = \zeta_{Z1}$. □

Remark 3.8 (Logical status of route B). Corollaries 3.3 and 3.7 give $\mathrm{(D)} \Rightarrow \mathrm{B}(C_\eta, M_*)$ and $[\mathrm{B}(C, M)$ and $\mathrm{U}(C, M)] \Rightarrow \mathrm{(D)}$. Route B therefore consists of two independent pieces. The first is existence, B, which is not weaker than (D). The second is uniqueness without the third condition of $\mathscr{O}$, Hypothesis 3.5, which does not involve ζ. Compactness itself is not an issue in the box: Proposition 3.4 uses only the imposed bounds. In this scheme the ordinates γ_k never enter the construction; they appear only as the unique admissible zero of $\mathcal{R}$. Lemma 3.2 does *not* show that near-solutions of sections lie in $\mathsf{K}(C_\eta, M_*)$: it locates only the exact solution. Restricting the sections to that box is a choice, justified a posteriori by Corollary 3.3.

3.1. **Route B with prescribed von Mangoldt masses.** The masses $\mathsf{a}_n^\zeta = \Lambda(n)/(2\sqrt{n})$ are known and do not depend on (D) or on the ordinates of ζ. Prescribing them leaves the ordinates as the only unknowns, and the uniqueness question of Hypothesis 3.5 disappears. We use the list form of `thm:distributional-EF` in the following form.

Hypothesis 3.9 (Convention (L))**.** For every C-admissible list $\boldsymbol{\kappa}$, the function $\widehat{\mathsf{R}}$ defined by (4.1) with the masses a^ζ (all n) is, on $\Re s > 1$, the Laplace transform of a distribution of finite order on $[0,\infty)$ whose restriction to $(0,\infty)$ is $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta)$.

With the masses a^ζ, $\sum_n \mathsf{a}_n^\zeta n^{\frac12 - s} = -\frac12 \frac{\zeta'}{\zeta}(s)$ on $\Re s > 1$, so

$$\widehat{\mathsf{R}} = \tfrac12 \Big(\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}} - \frac{\xi'}{\xi} \Big), \qquad \xi := \tfrac12 G\zeta. \tag{3.2}$$

Hypothesis 3.9 is the statement that (3.2) is the Laplace transform of the residual distribution of (2.4). It has to be checked against `thm:distributional-EF` and is not proved here.

Proposition 3.10 (Uniqueness with prescribed masses)**.** *Assume Hypothesis 3.9. Let $\boldsymbol{\kappa}$ be C-admissible for some C, and suppose $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta) = 0$ in $\mathcal{D}'(0,\infty)$. Then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, and $\boldsymbol{\kappa} = \boldsymbol{\gamma}$.*

Proof. By Hypothesis 3.9 the distribution with Laplace transform $\widehat{\mathsf{R}}$ is supported at $\{0\}$ and has finite order, so $\widehat{\mathsf{R}}$ is a polynomial p. By (3.2), $(P_{\boldsymbol{\kappa}} e^{-Q}/\xi)' = 0$ on $\Re s > 1$ with $Q' = 2p$, Q a polynomial. Hence $P_{\boldsymbol{\kappa}} = c\,\xi\, e^{Q}$ there with $c \neq 0$, and by analytic continuation on all of $\mathbb{C}$. In the variable $w = s - \frac{1}{2}$, $P_{\boldsymbol{\kappa}}$ is a constant multiple of $\prod_k (1 + w^2/\kappa_k^2)$, and $N_{\boldsymbol{\kappa}}(t) \ll t \log t$, so $P_{\boldsymbol{\kappa}}$ has order 1; ξ has order 1. Thus $e^{Q} = P_{\boldsymbol{\kappa}}/(c\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \leq 1$ by Hadamard's theorem. Since $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$ and $\xi(1-s) = \xi(s)$, $e^{Q(1-s)} = e^{Q(s)}$; for $Q(s) = a + bs$ this forces $b(1-2s) \in 2\pi i\mathbb{Z}$ for all s, so $b = 0$. Therefore $P_{\boldsymbol{\kappa}} = c'\xi$. The zeros of $P_{\boldsymbol{\kappa}}$ are $\frac{1}{2} \pm i\kappa_k$, so every zero of ξ, that is, every nontrivial zero of ζ, has real part $\frac{1}{2}$, and the ordinates counted with multiplicity are the κ_k. Both lists are nondecreasing, so $\boldsymbol{\kappa} = \boldsymbol{\gamma}$. □

Definition 3.11 (Statement B′)**.** *Statement $B'(C)$* says that there are C-admissible lists $\boldsymbol{\kappa}^{(m)}$ with $\kappa_k^{(m)} = \tau_k$ for $k > K_m$ and $\rho_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta) \to 0$.

Corollary 3.12 (Route B′)**.** *Assume Hypothesis 3.9.*

(a) *If $B'(C)$ holds for some C, then every sequence as in Definition 3.11 converges coordinatewise to $\boldsymbol{\gamma}$, and all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$.*
(b) *If all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, then $B'(C)$ holds for every $C \geq C_\eta$.*

Proof. (a) $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta) \in \mathsf{K}(C, M_*)$ by Lemma 3.2. By Proposition 3.4, every subsequence has a further subsequence converging coordinatewise to some $(\boldsymbol{\kappa}, \mathsf{a}^\zeta)$ with $\mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta) = 0$ and $\boldsymbol{\kappa}$ C-admissible; the masses do not move. By Proposition 3.10, $\boldsymbol{\kappa} = \boldsymbol{\gamma}$ and the zeros lie on the line. Since every subsequence has a further subsequence with the same limit, the whole sequence converges. (b) Under this hypothesis the rung system with $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$ is the distributional explicit formula `thm:distributional-EF`, and $|\gamma_k - \tau_k| \leq C_\eta$ by `lem:bounded-displacement`. The truncations in the proof of Corollary 3.3, with the masses kept equal to a^ζ, give the sections. □

Remark 3.13 (What prescribing the masses changes). Under Hypothesis 3.9, Hypothesis 3.5 is no longer needed: uniqueness holds for every list at bounded distance from the clock, not only in $\mathscr{O}$, and it follows from Hadamard's theorem and the functional equation instead of `thm:free-masses`. Route B′ has a single open piece, the existence statement B′, which by Corollary 3.12 is equivalent to the Riemann hypothesis. With prescribed masses the whole window $(0, U)$ constrains the ordinates; the part above $\log \Omega$ can no longer be absorbed by the masses, as it could in Heuristic 4.12.

3.2. **Staged construction and the extension lemma.**

Definition 3.14 (Staged sections)**.** Fix $U_1 < U_2 < \cdots \to \infty$, test sets Φ_m supported in $(0, U_m)$ and nested as in Section 3, and $K_m \to \infty$. Stage m has unknowns $\kappa_1, \ldots, \kappa_{K_m}$, with $\kappa_k = \tau_k$ for $k > K_m$ and masses a^ζ; it is started at $\kappa_k^{(m,0)} = \kappa_k^{(m-1)}$ for $k \leq K_{m-1}$ and $\kappa_k^{(m,0)} = \tau_k$ for $K_{m-1} < k \leq K_m$, with $\kappa^{(0)} = \boldsymbol{\tau}$. Its residual vector is $R_m(\boldsymbol{\kappa}) := (\langle \mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta), f\rangle)_{f \in \Phi_m}$, and its Jacobian is

$$(J_m)_{f,k} = \partial_{\kappa_k} \langle \mathcal{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta), f\rangle = -\int_0^\infty f(u)\, u \sin(\kappa_k u)\, du \qquad (f \in \Phi_m,\ k \leq K_m). \tag{3.3}$$

A Gauss–Newton step is $h = -J_m^\dagger R_m$, $\boldsymbol{\kappa} \leftarrow \boldsymbol{\kappa} + h$.

Hypothesis 3.15 (Extension lemma)**.** There are $C > 0$, $q \in (0,1)$, $A > 0$, $r_m \to \infty$ and $\eta_m \geq 0$ with $\sum_m \eta_m < \infty$ such that the staged sections can be chosen C-admissible with $E_m := \rho_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{\zeta})$ satisfying

$$E_{m+1} \leq qE_m + \eta_m, \qquad \max_{k \leq r_m} |\kappa_k^{(m+1)} - \kappa_k^{(m)}| \leq A\, E_m. \tag{3.4}$$

Proposition 3.16 (Extension implies route B′)**.** *Hypothesis 3.15 implies $E_m \to 0$ and $\sum_m E_m < \infty$. Hence each coordinate $\kappa_k^{(m)}$ is a Cauchy sequence, $B'(C)$ holds, and under Hypothesis 3.9 the limit is $\boldsymbol{\gamma}$ and the Riemann hypothesis holds.*

Proof. Iterating, $E_m \leq q^{m-1}E_1 + \sum_{j<m} q^{m-1-j}\eta_j$, and $\sum_m E_m \leq \frac{E_1}{1-q} + \frac{1}{1-q}\sum_j \eta_j < \infty$. For fixed k and m with $r_m \geq k$, $|\kappa_k^{(m+1)} - \kappa_k^{(m)}| \leq AE_m$, which is summable. The rest is Corollary 3.12(a). □

Remark 3.17 (Status of the extension lemma)*.* By Proposition 3.16, Hypothesis 3.15 implies the Riemann hypothesis; it is not a weaker intermediate step. Its content is a stability statement for the linearised equation $-\sum_k h_k\, u \sin(\kappa_k u) \simeq D_m(u)$ behind (3.3). A bound $|h_k| \leq C_k \|R_m\|$ for $k \leq r$ and $m \geq m(r)$, with C_k independent of m, is what a proof of (3.4) would need. A bound with C_k independent of k is a uniform stability statement of the kind ruled out, for the unfolded zeta list on the whole line, by Theorem 6.8; this connection is heuristic, since Theorem 6.8 concerns the Riesz-basis property in the unfolded variable and not the matrices J_m. Constants C_k that grow with k are not excluded by anything proved here. Injectivity of J_m at the clock, needed to start the Gauss–Newton iteration, is open (`lem:cauchy-jacobian`).

4. An a priori bound for quasi-solutions

Throughout this section $(\boldsymbol{\kappa}, \mathsf{a})$ is a finite section in $\mathsf{K}(C, M)$ with parameters K, N.

Definition 4.1 (Laplace residual)**.** For $\Re s > 1$ put

$$\widehat{\mathsf{R}}(s) := \tfrac{1}{2}\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s) - \tfrac{1}{2}\frac{G'}{G}(s) + \sum_{n=2}^{N} \mathsf{a}_n\, n^{\frac{1}{2}-s}. \tag{4.1}$$

For $T \geq 0$ put

$$E(T) := \int_{-T}^{T} |\widehat{\mathsf{R}}(2+iv)|\, dv, \qquad \varepsilon := \sup_{T \geq 2} \frac{E(T)}{\log T} \in [0, \infty]. \tag{4.2}$$

Since $\int_0^\infty e^{-(s-\frac{1}{2})u}\cos(\kappa u)\, du = \frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2+\kappa^2} = \frac{1}{2}\frac{d}{ds}\log(1 + \frac{q(s)}{\frac{1}{4}+\kappa^2})$ and $\int_0^\infty e^{-(s-\frac{1}{2})u}\delta(u - \log n)\, du = n^{\frac{1}{2}-s}$, $\widehat{\mathsf{R}}$ is the Laplace transform of the residual distribution of (2.4), with the normalisation of k fixed in `eq:transport-defs`. We take (4.1) as the definition. Because the coefficients are real, $\widehat{\mathsf{R}}(\bar{s}) = \overline{\widehat{\mathsf{R}}(s)}$, so $E(T) = 2\int_0^T |\widehat{\mathsf{R}}(2+iv)|\, dv$.

Constants. Put $\ell_0 := 2\sum_{n \geq 2} n^{-3/2}/\log n < \infty$, and, for $s \in \mathbb{C}$ and real $x \geq 2\pi$, $f(x; s) := \log(1 + \frac{q(s)}{\frac{1}{4}+x^2})$, with the principal branch, which is defined whenever $|s - \frac{1}{2}| < x$. Let $\mathscr{S} := \{2\} \cup \{\sigma + i : \ -1 \leq \sigma \leq 2\}$, a compact set on which $|s - \frac{1}{2}| \leq \frac{5}{2} < 2\pi$, and

$$\beta(C) := \sum_{k \geq 1} \sup_{x \in [\max(\tau_k - C, 2\pi),\, \tau_k + C]} \sup_{s \in \mathscr{S}} |f(x; s) - f(\tau_k; s)|. \tag{4.3}$$

This is finite. On $\mathscr{S}$, $|q(s)| \leq 5$ and $|\partial_x f(x; s)| = |\frac{2xq(s)}{(\frac{1}{4}+x^2)(x^2+(s-\frac{1}{2})^2)}| \leq \frac{20}{x^3}$ for $x \geq 4$. The k-th term is therefore at most $160C\tau_k^{-3}$ when $\tau_k \geq 2C$, and $\sum_k \tau_k^{-3} < \infty$. Further, $c_\tau := \max_{-1 \leq \sigma \leq 2} |\log|\zeta_{\mathrm{RVM}}(\sigma + i)||$ and $c_{\mathrm{RVM}} := \sup_{\Re s \geq 2} |\log \zeta_{\mathrm{RVM}}(s)|$. Both are finite: the first because ζ_{RVM} has no zeros or poles on that segment, the second by `thm:halfplane-RVM`.

Lemma 4.2 (Phragmén–Lindelöf, two forms)**.** (a) *Let* $\Sigma := \{\alpha \le \Re s \le \beta,\ \Im s \ge t_1\}$ *have width* $a := \beta - \alpha$. *Let* u *be upper semicontinuous on* Σ *and subharmonic in its interior, with* $u \le M_0$ *on* $\partial\Sigma$ *and* $u(s) \le A_0 e^{c\Im s}$ *for some* $c < \pi/a$. *Then* $u \le M_0$ *on* Σ.
(b) *Let* g *be holomorphic in* $\Re s > 1$, *continuous on* $\Re s \ge 2$, *with* $|g| \le 1$ *on* $\Re s = 2$ *and* $|g(s)| \le A_0(2+|s|)^L$ *on* $\Re s \ge 2$. *Then* $|g| \le 1$ *on* $\Re s \ge 2$.

Proof. (a) Choose $c < c' < \pi/a$ and $m := \frac{\alpha+\beta}{2}$, and put $\omega(s) := e^{c'\Im s}\cos(c'(\Re s - m))$. This is harmonic, being the real part of $e^{-ic'(s-m)}$. On Σ, $\omega \ge \cos(c'a/2)\, e^{c'\Im s}$, and $\cos(c'a/2) > 0$. For $\eta > 0$, $u - M_0 - \eta\omega$ is subharmonic, ≤ 0 on $\partial\Sigma$, and tends to $-\infty$ as $\Im s \to \infty$. The maximum principle on $\Sigma \cap \{\Im s \le t_2\}$, with t_2 large, gives $u \le M_0 + \eta\omega$. Let $\eta \to 0$.
(b) For $\eta > 0$, $g_\eta(s) := g(s)\exp(-\eta(s-1)^{1/2})$, with the principal branch, satisfies $|g_\eta| \le |g|$ on $\Re s \ge 2$, because $\Re(s-1)^{1/2} \ge \cos(\pi/4)|s-1|^{1/2} > 0$ there. Moreover $g_\eta(s) \to 0$ as $|s| \to \infty$ in $\Re s \ge 2$. The maximum principle on large half-discs gives $|g_\eta| \le 1$. Let $\eta \to 0$. □

Lemma 4.3 (The Dirichlet factor)**.** *Put* $\mathcal{L}_{\mathsf{a}}(s) := \sum_{n=2}^{N} \frac{2\mathsf{a}_n}{\log n} n^{\frac12 - s}$ *and* $Y := \zeta_{\boldsymbol\kappa} e^{-\mathcal{L}_{\mathsf{a}}}$. *Then* Y *is holomorphic and zero-free in* $\Re s > 1$, *and* $|\mathcal{L}_{\mathsf{a}}| \le M\ell_0$ *on* $\Re s \ge 2$. *For the branch of* $\log Y$ *that is real on* $(1,\infty)$,

$$(\log Y)' = 2\,\widehat{\mathsf{R}}, \qquad |\log Y(2)| \le B_0 := |\log \zeta_{\mathrm{RVM}}(2)| + \beta(C) + M\ell_0, \tag{4.4}$$

and, for all real t,

$$|\log Y(2+it)| \le B_0 + E(|t|) \le B_0 + \varepsilon\log(2+|t|). \tag{4.5}$$

Proof. $P_{\boldsymbol\kappa}$ vanishes only on $\Re s = \frac12$, and G is holomorphic and zero-free in $\Re s > 1$. Hence $\zeta_{\boldsymbol\kappa}$ and Y are holomorphic and zero-free there. $|\mathcal{L}_{\mathsf{a}}(s)| \le \sum_n 2Mn^{\frac12 - \Re s}/\log n \le M\ell_0$ for $\Re s \ge 2$. Differentiating, $(\log Y)' = \frac{P'_{\boldsymbol\kappa}}{P_{\boldsymbol\kappa}} - \frac{G'}{G} + 2\sum_n \mathsf{a}_n n^{\frac12 - s} = 2\widehat{\mathsf{R}}$. At $s = 2$, $\log\zeta_{\boldsymbol\kappa}(2) = \log\zeta_{\mathrm{RVM}}(2) + \sum_{k\le K}(f(\kappa_k;2) - f(\tau_k;2))$, a finite sum of real numbers bounded by $\beta(C)$, and $|\mathcal{L}_{\mathsf{a}}(2)| \le M\ell_0$. Finally, $\log Y(2+it) - \log Y(2) = 2i\int_0^t \widehat{\mathsf{R}}(2+iv)\,dv$ has modulus at most $E(|t|)$. For $|t| \ge 2$, $E(|t|) \le \varepsilon\log|t|$; for $|t| < 2$, $E(|t|) \le E(2) \le \varepsilon\log 2$. □

Lemma 4.4 (Bound on the half-plane)**.** *With* $W := \log|\zeta_{\boldsymbol\kappa}| = \log|P_{\boldsymbol\kappa}| - \log|G|$ *and* $B_1 := B_0 + \varepsilon\log(2\sqrt2)$, $B_2 := B_1 + M\ell_0$, *for* $\Re s \ge 2$,

$$\big|\log|Y(s)|\big| \le B_1 + \varepsilon\log|s-1|, \qquad |W(s)| \le B_2 + \varepsilon\log|s-1|. \tag{4.6}$$

Proof. On $\Re s = 2$, $|s-1| = \sqrt{1+t^2} \ge (2+|t|)/(2\sqrt2)$, so (4.5) gives $|Y(s)|^{\pm1} \le e^{B_1}|s-1|^{\varepsilon}$. A priori growth: on $\Re s \ge 2$, $|\log|\zeta_{\mathrm{RVM}}|| \le c_{\mathrm{RVM}}$. The quotient $\zeta_{\boldsymbol\kappa}/\zeta_{\mathrm{RVM}}$ is a finite product of K factors $\frac{(\kappa_k^2+w^2)(\frac14+\tau_k^2)}{(\frac14+\kappa_k^2)(\tau_k^2+w^2)}$ with $w = s - \frac12$. Since $|\kappa^2+w^2| = |\kappa - \Im s + i\Re w|\,|\kappa + \Im s - i\Re w| \ge (\Re w)^2 \ge \frac94$, each factor and its inverse are $O((2+|s|)^4)$. Hence $|Y|^{\pm1} \le A_0(2+|s|)^{4K}$. Apply Lemma 4.2(b) to $g = Y^{\pm1}(s-1)^{-\varepsilon}e^{-B_1}$, which is holomorphic in $\Re s > 1$. Finally $W = \log|Y| + \Re\mathcal{L}_{\mathsf{a}}$. □

Lemma 4.5 (Bound on the half-strip)**.** *Let* $\Sigma := \{-1 \le \Re s \le 2,\ \Im s \ge 1\}$ *and* $B_3 := \max\big(B_2 + \frac34\log 2,\ c_\tau + \beta(C)\big)$. *Then*

$$W(s) \le B_3 + \big(\tfrac32 + \varepsilon\big)\log|s+2| \qquad (s \in \Sigma). \tag{4.7}$$

Proof. W is subharmonic in the interior of Σ, since $\log|G|$ is harmonic there (G has no zeros or poles with $\Im s \ge 1$), and it is upper semicontinuous on Σ. The majorant $h(s) := (\frac32 + \varepsilon)\log|s+2|$ is harmonic and nonnegative on Σ. We check the three sides.

Right side. By (4.6), $W(2+it) \le B_2 + \varepsilon\log|1+it| \le B_2 + h(2+it)$.

Left side. $P_{\boldsymbol\kappa}$ and G are real, so $W(\bar s) = W(s)$. Combined with $P_{\boldsymbol\kappa}(1-s) = P_{\boldsymbol\kappa}(s)$, this gives $W(-1+it) = W(2+it) + \log|G(2+it)/G(-1+it)|$. By $\Gamma(z+1) = z\Gamma(z)$ and the reflection

formula, $|\Gamma(\frac12+iy)|^2=\pi/\cosh\pi y$ and $|\Gamma(1+iy)|^2=\pi y/\sinh\pi y$, so

$$\Big|\frac{G(2+it)}{G(-1+it)}\Big|=\frac{|1+it|}{2}\,\pi^{-3/2}\Big(\frac{t}{2}\coth\frac{\pi t}{2}\Big)^{1/2}\le(1+t)^{3/2}\qquad(t\ge1).\tag{4.8}$$

For the inequality: $|1+it|\le1+t$, $\frac{t}{2}\coth\frac{\pi t}{2}\le\frac{1+t}{2}\coth\frac{\pi}{2}$, and $\frac12\pi^{-3/2}(\frac12\coth\frac{\pi}{2})^{1/2}<1$. Since $(1+t)\le\sqrt{2}|1+it|$ and $|s+2|=|1+it|$ at $s=-1+it$, we get $W(-1+it)\le B_2+\frac34\log2+h(-1+it)$.

Bottom side. For $s=\sigma+i$, $W(s)=\log|\zeta_{\mathrm{RVM}}(s)|+\sum_{k\le K}\Re(f(\kappa_k;s)-f(\tau_k;s))\le c_\tau+\beta(C)\le c_\tau+\beta(C)+h(s)$.

Growth. $\log|P_{\boldsymbol\kappa}(s)|\le\sum_k\log(1+|q(s)|/\kappa_k^2)=O(|s|\log^2|s|)$, and $-\log|G(s)|\le\frac{\pi}{4}|\Im s|+O(\log|s|)$ on Σ by Stirling's formula [12, §12.33]. Hence $W-h\le A_0e^{c\Im s}$ for any $c>0$. Lemma 4.2(a), with width 3 and $c<\pi/3$, applied to $W-h$ gives the claim. □

Lemma 4.6 (Backlund–Jensen on a horizontal segment)**.** *There are absolute constants $T_0\ge3$ and c_4 such that, for $T\ge T_0$ with $T\notin\{\kappa_k\}$, the variation of $\arg P_{\boldsymbol\kappa}$ along the segment from $2+iT$ to $\frac12+iT$ is at most $\pi(\nu+1)$, where*

$$\nu\le\frac{(\frac52+\varepsilon)\log(T+8)+E(T)+B_0+B_3+M\ell_0+2c_4}{\log(4/3)}.\tag{4.9}$$

Proof. Choose $\alpha\in\mathbb{R}$ with $e^{i\alpha}P_{\boldsymbol\kappa}(2+iT)>0$ and put $F_T(z):=\frac12(e^{i\alpha}P_{\boldsymbol\kappa}(z+iT)+e^{-i\alpha}P_{\boldsymbol\kappa}(z-iT))$, which is entire. Since $P_{\boldsymbol\kappa}$ is real, $P_{\boldsymbol\kappa}(x-iT)=\overline{P_{\boldsymbol\kappa}(x+iT)}$ for real x, so $F_T(x)=\Re(e^{i\alpha}P_{\boldsymbol\kappa}(x+iT))$ and $F_T(2)=|P_{\boldsymbol\kappa}(2+iT)|$. If F_T has ν zeros on $[\frac12,2]$, the segment splits into at most $\nu+1$ pieces on which $\Re(e^{i\alpha}P_{\boldsymbol\kappa})$ has constant sign. On each piece arg varies by less than π, which gives the first claim [11, §9.4].

Jensen's formula on $|z-2|\le R=2$, which contains the disc $|z-2|\le r=\frac32$ and hence $[\frac12,2]$, gives $\nu\log\frac{R}{r}\le\max_{|z-2|=2}\log|F_T(z)|-\log|F_T(2)|$. Moreover $|F_T(z)|\le\max_{|\zeta-2|\le2}|P_{\boldsymbol\kappa}(\zeta+iT)|$, because $|P_{\boldsymbol\kappa}(z-iT)|=|P_{\boldsymbol\kappa}(\bar z+iT)|$. Write $\log|P_{\boldsymbol\kappa}|=W+\log|G|$. On the disc $\{|s-(2+iT)|\le2\}$, the part with $\Re s\le2$ lies in Σ and $|s+2|\le T+6$, so Lemma 4.5 gives $W\le B_3+(\frac32+\varepsilon)\log(T+6)$. The part with $\Re s\ge2$ satisfies $W\le B_2+\varepsilon\log(T+5)$ by Lemma 4.4, a smaller bound. At the centre, (4.5) gives $W(2+iT)\ge-(B_0+E(T)+M\ell_0)$. For the oscillation of $\log|G|$ on the disc: by Stirling's formula for Γ'/Γ in $|\arg z|\le\frac34\pi$ [12, §12.33], $|G'/G(s)|=|\frac1s+\frac{1}{s-1}-\frac12\log\pi+\frac12\frac{\Gamma'}{\Gamma}(\frac{s}{2})|\le\frac12\log(T+8)+c_4$ on the disc for $T\ge T_0$. So the oscillation is at most $\log(T+8)+2c_4$. Collecting terms gives (4.9). □

Lemma 4.7 (Argument principle)**.** *For $T>0$, $T\notin\{\kappa_k\}$, let Γ_T be the path $2\to2+iT\to\frac12+iT$. Then*

$$\pi N_{\boldsymbol\kappa}(T)=\Delta_{\Gamma_T}\arg P_{\boldsymbol\kappa}=\pi+\vartheta(T)+\Delta_{\Gamma_T}\arg\zeta_{\boldsymbol\kappa},\tag{4.10}$$

where $\vartheta(T)=\arg(\pi^{-iT/2}\Gamma(\frac14+\frac{iT}{2}))$ is the Riemann–Siegel theta function, defined by continuous variation from $\vartheta(0)=0$.

Proof. Take the rectangle $\mathcal{B}$ with vertices $2\pm iT$, $-1\pm iT$. $P_{\boldsymbol\kappa}$ has no zeros on $\partial\mathcal{B}$: its zeros lie on $\Re s=\frac12$, and $T\notin\{\kappa_k\}$. Its zeros inside $\mathcal{B}$ are $\frac12\pm i\kappa_k$ with $\kappa_k<T$, $2N_{\boldsymbol\kappa}(T)$ in total. Reflection $s\mapsto1-\bar s$ maps Γ_T to the path $-1\to-1+iT\to\frac12+iT$. Since $P_{\boldsymbol\kappa}(1-\bar s)=\overline{P_{\boldsymbol\kappa}(s)}$, the variation of $\arg P_{\boldsymbol\kappa}$ along the reversed image path $\frac12+iT\to-1+iT\to-1$ equals $\Delta_{\Gamma_T}\arg P_{\boldsymbol\kappa}$. The two lower quarters are handled in the same way with $s\mapsto\bar s$. The four quarter paths make up $\partial\mathcal{B}$, so $4\Delta_{\Gamma_T}\arg P_{\boldsymbol\kappa}=2\pi\cdot2N_{\boldsymbol\kappa}(T)$.

For the second equality, write $P_{\boldsymbol\kappa}=G\zeta_{\boldsymbol\kappa}$. Along Γ_T, $s(s-1)$ moves from 2 to $-(\frac14+T^2)$ inside the closed upper half-plane, so its argument varies by π. $\pi^{-s/2}\Gamma(s/2)$ is holomorphic and zero-free in $\Re s>0$, so its argument variation along Γ_T equals that along $[2,\frac12]\cup[\frac12,\frac12+iT]$, which is $0+\vartheta(T)$. □

Theorem 4.8 (Counting bound for quasi-solutions)**.** *There are absolute constants T_0 and a_4 such that, for every finite section in $\mathsf{K}(C,M)$ and every $T \ge T_0$,*

$$\bigl|N_{\boldsymbol{\kappa}}(T) - N_\tau(T)\bigr| \le (a_1 + a_2\varepsilon)\log(T+8) + a_3 B + a_4, \tag{4.11}$$

with

$$a_1 = \frac{5}{2\log(4/3)} = 8.690\ldots, \quad a_2 = \frac{2}{\log(4/3)} + \frac{1}{\pi} = 7.270\ldots, \quad a_3 = \frac{1}{\log(4/3)} + \frac{1}{\pi} = 3.794\ldots, \tag{4.12}$$

and $B := B_0 + B_3 + M\ell_0$. The constants $a_1, \ldots, a_4$ do not depend on K, N, C, M or ε; B depends on C (through $\beta(C)$), on M and on ε.

Proof. By continuity from the right it suffices to take $T \notin \{\kappa_k\}$. By Lemma 4.7, $N_{\boldsymbol{\kappa}}(T) - 1 - \vartheta(T)/\pi = \pi^{-1}\Delta_{\Gamma_T}\arg\zeta_{\boldsymbol{\kappa}}$.

Vertical piece. $\arg\zeta_{\boldsymbol{\kappa}} = \Im\log Y + \Im\mathcal{L}_{\mathsf{a}}$, with $\Im\log Y(2) = \Im\mathcal{L}_{\mathsf{a}}(2) = 0$. Its variation is at most $E(T) + M\ell_0$ by Lemma 4.3.

Horizontal piece. The variation is $\Delta\arg P_{\boldsymbol{\kappa}} - \Delta\arg G$. The first is at most $\pi(\nu+1)$ by Lemma 4.6. The second is $-\Im\int_{1/2}^{2}(G'/G)(\sigma+iT)\,d\sigma$. Here $|\Im(G'/G)| \le |\frac{1}{s}| + |\frac{1}{s-1}| + \frac{1}{2}|\arg\frac{s}{2}| + O(1/T) \le 1$ for $T \ge T_0$, by Stirling's formula for Γ'/Γ [12, §12.33]. So the second variation is at most $c_5 := \frac{3}{2}$.

Main term. By the Stirling expansion of ϑ [12, §12.33], $1 + \vartheta(T)/\pi = \mathcal{N}_{\mathrm{RVM}}(T) + O(1/T)$, and this error is at most $1/T$ for $T \ge T_0$. With (2.2):

$$|N_{\boldsymbol{\kappa}}(T) - N_\tau(T)| \le \nu + 1 + \frac{E(T) + M\ell_0 + c_5}{\pi} + \frac{1}{T} + \frac{1}{2}. \tag{4.13}$$

Insert (4.9) and $E(T) \le \varepsilon\log(T+8)$. □

Corollary 4.9 (Displacement bound)**.** *Assume $T_0 \ge 2\pi e$. For every k with $\tau_k - C \ge T_0$,*

$$|\kappa_k - \tau_k| \le \frac{2\pi\bigl[(a_1 + a_2\varepsilon)\log(\tau_k + C + 8) + a_3 B + a_4\bigr]}{\log\bigl((\tau_k - C)/2\pi\bigr)}. \tag{4.14}$$

In particular $\limsup_{k\to\infty}|\kappa_k - \tau_k| \le 2\pi(a_1 + a_2\varepsilon)$ uniformly over sections with given C, M, ε. For every $\eta > 0$ there is $k_0(C, M, \varepsilon, \eta)$ such that $|\kappa_k - \tau_k| \le 2\pi(a_1 + a_2\varepsilon) + \eta$ for $k \ge k_0$.

Proof. Let $\Phi(T)$ be the right-hand side of (4.11); it is nondecreasing. Suppose $\kappa_k > \tau_k + d$ with $0 < d \le C$, and let $T := \tau_k + d - \eta'$ with small $\eta' > 0$. The list is nondecreasing, so $N_{\boldsymbol{\kappa}}(T) \le k - 1$. By (2.2) and because $\mathcal{N}'_{\mathrm{RVM}}$ is increasing, $N_\tau(T) \ge \mathcal{N}_{\mathrm{RVM}}(T) - \frac{1}{2} \ge k - 1 + (d - \eta')\mathcal{N}'_{\mathrm{RVM}}(\tau_k)$. Hence $(d-\eta')\frac{1}{2\pi}\log\frac{\tau_k}{2\pi} \le \Phi(\tau_k + C)$. Now suppose $\kappa_k < \tau_k - d$, and let $T := \tau_k - d + \eta'$. Then $N_{\boldsymbol{\kappa}}(T) \ge k$ and $N_\tau(T) \le \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \le k - (d-\eta')\mathcal{N}'_{\mathrm{RVM}}(\tau_k - C)$. Let $\eta' \to 0$ in both cases. □

Remark 4.10 (What Corollary 4.9 does not give). The proof uses the imposed tube $|\kappa_k - \tau_k| \le C$ twice: through $\beta(C)$ in B, and to replace $\Phi(\tau_k + d)$ by $\Phi(\tau_k + C)$. The conclusion is that, inside any finite tube, the coordinates of sufficiently high index lie in the smaller tube of radius $2\pi(a_1 + a_2\varepsilon) + \eta$. So the constraint is inactive there once $C > 2\pi(a_1 + a_2\varepsilon) + \eta$. For the indices $k < k_0(C, M, \varepsilon, \eta)$ nothing is proved, and the corollary does not show that unconstrained near-minimisers lie in a universal tube. The proof uses finite sections: Lemma 4.4 needs the polynomial growth $A_0(2 + |s|)^{4K}$. Extending it to infinite lists requires a separate growth bound for $P_{\boldsymbol{\kappa}}/P_{\boldsymbol{\tau}}$ on $\Re s \ge 2$, which is not given here. The constant $2\pi a_1 = 54.60\ldots$ is far from optimal.

Heuristic 4.11 (What a finite section controls)**.** The quantity in Theorem 4.8 is $E(T)$, the L^1 norm on $\Re s = 2$ of the Laplace transform of the residual. It is not the Euclidean residual that a numerical section minimises over finitely many tests. Let the tests be $w(u)\varphi(u)$ with a fixed window $w \in C_c^\infty(0,\infty)$, $w \equiv 1$ on $[u_0, U]$, and φ band-limited to $|\omega| \le \Omega$. Then the kernel $w(u)e^{-\frac{3}{2}u}e^{-ivu}$ is, approximately, in the span of the tests only for $|v| \lesssim \Omega$. The residual on $(0, u_0)$ is then not controlled, and for a list at distance C from the clock it is expected to be of size $C\log(1/u)$ as

$u \to 0$. Its contribution to $E(T)$ is then $O(CTu_0 \log(1/u_0))$. Since $u_0 \gtrsim 1/\Omega$ in a band-limited section, Theorem 4.8 would apply with a given ε_0 only at heights $T \lesssim \varepsilon_0 \Omega/(C \log \Omega)$. None of these size estimates is proved here. They indicate which norm a section must make small. Lemma 4.3 shows that this norm is the right one.

Heuristic 4.12 (Informative part of a section)**.** Where the atoms are denser than the resolution of the tests, $e^{-u} \ll \pi/\Omega$, the masses alone can approximate a smooth target by a Riemann sum $\mathsf{a}_n \approx g(\log n)/n$ without moving any ordinate. That part of the window then carries no information on the ordinates. Counting unknowns against equations in the remaining window, $U \le \log \Omega + O(1)$, gives an overdetermined system. This proves nothing about compatibility. It only explains why B is an existence statement and not a dimension count.

5. A Kadec–Avdonin criterion with blocks of variable length

5.1. **Discrepancy and generating function.**

Definition 5.1 (Discrepancy, Poisson and conjugate operators)**.** Let $X = (x_k)_{k\in\mathbb{Z}}$ be a real non-decreasing sequence with

$$|x_k - (k - \tfrac{1}{2})| \le c_X \log(2 + |k|) \qquad (k \in \mathbb{Z}). \tag{5.1}$$

Its *discrepancy* with respect to the half-integers is

$$\psi_X(x) := \sum_{k\in\mathbb{Z}} \big(\mathbf{1}_{[x_k,\infty)}(x) - \mathbf{1}_{[k-\frac{1}{2},\infty)}(x)\big). \tag{5.2}$$

The k-th term is ± 1 on the interval between x_k and $k - \frac{1}{2}$ and 0 elsewhere. By (5.1), at most $O(\log(2+|x|))$ terms are nonzero at x, so $|\psi_X(x)| \le c'_X \log(2+|x|)$ and $\psi_X \in L^1(dx/(1+x^2))$. For $g \in L^1(dx/(1+x^2))$ and $y > 0$ put

$$P_y g(x) := \frac{1}{\pi} \int \frac{y\, g(s)\, ds}{(x-s)^2 + y^2}, \qquad \mathcal{Q}_y g(x) := \frac{1}{\pi} \int g(s) \Big[\frac{x-s}{(x-s)^2+y^2} + \frac{s}{1+s^2}\Big] ds. \tag{5.3}$$

The last term in $\mathcal{Q}_y$ only makes the integral converge. Changing it changes $\mathcal{Q}_y g$ by a constant, which is irrelevant for all statements below.

Lemma 5.2 (Generating function)**.** *Let X satisfy* (5.1)*, let $F_0(z) := \cos \pi z$, and fix $z_0 \in \mathbb{R} \setminus (X \cup (\mathbb{Z} + \frac{1}{2}))$. The series*

$$H_X(z) := \sum_{k\in\mathbb{Z}} \Big(\frac{1}{z - x_k} - \frac{1}{z - k + \frac{1}{2}} \Big) \tag{5.4}$$

converges absolutely and locally uniformly off $X \cup (\mathbb{Z} + \frac{1}{2})$, and

$$F_X(z) := F_0(z) \exp \int_{z_0}^{z} H_X(\zeta)\, d\zeta \tag{5.5}$$

is a real entire function whose zero set is X, with multiplicities. Moreover:

(i) *for $y > 0$ there is a constant c''_X such that*

$$\log |F_X(x+iy)| = \log |F_0(x+iy)| + \pi\, \mathcal{Q}_y \psi_X(x) + c''_X; \tag{5.6}$$

(ii) *F_X is of exponential type π, and its indicator diagram is the segment $[-i\pi, i\pi]$;*

(iii) *if X is symmetric, $x_{1-k} = -x_k$, and $z_0 = 0 \notin X$, then F_X is even.*

Proof. Each term of H_X equals $\frac{x_k - k + \frac{1}{2}}{(z-x_k)(z-k+\frac{1}{2})} = O(\log(2+|k|)/k^2)$ locally uniformly, so the series converges. H_X is meromorphic with simple poles, with residue $+m$ at a point of X of multiplicity m and -1 at each half-integer; coincident poles combine their residues. The residues are integers, so $\exp \int_{z_0}^{z} H_X$ is single-valued. It has a pole of order one at each half-integer not in X, cancelled

by the zero of F_0. Hence F_X is entire with zero set X. $H_X(\bar z)=\overline{H_X(z)}$ and z_0 is real, so F_X is real on $\mathbb{R}$. Under the symmetry in (iii), H_X is odd, so $\int_0^z H_X$ is even, and F_X is even.

(i) For $\Im z\neq 0$ put $\Phi_X(z):=\int \psi_X(s)\big[\frac{1}{z-s}+\frac{s}{1+s^2}\big]ds$. For $f(s):=\frac{1}{z-s}$, the identity $f(x_k)-f(k-\frac12)=-\int f'(s)(\mathbf{1}_{[x_k,\infty)}-\mathbf{1}_{[k-\frac12,\infty)})(s)\,ds$ summed over k (Fubini applies because $\sum_k\int|f'||\cdots|<\infty$ by (5.1)) gives $H_X(z)=-\int\psi_X(s)(z-s)^{-2}ds=\Phi_X'(z)$. So $F_X=c_+F_0e^{\Phi_X}$ on $\Im z>0$, with $c_+\neq 0$. Since $\Re\frac{1}{z-s}=\frac{x-s}{(x-s)^2+y^2}$, $\Re\Phi_X(x+iy)=\pi\mathcal{Q}_y\psi_X(x)$ and $\Im\Phi_X(x+iy)=-\pi P_y\psi_X(x)$.

(ii) Split $\mathcal{Q}_y\psi_X(x)$ into $|s-x|\le 1$, $1<|s-x|$ with $|s|\le 2|z|+2$, and $|s|>2|z|+2$. Using $|\psi_X(s)|\le c_X'\log(2+|s|)$, $\int_{|u|\le1}\frac{|u|\,du}{u^2+y^2}=\log(1+y^{-2})$, and, for $|s|>2|z|+2$, $|\frac{x-s}{(x-s)^2+y^2}+\frac{s}{1+s^2}|\le 8(1+|z|)/s^2$, one gets

$$|\mathcal{Q}_y\psi_X(x)|\le c_X'''(1+\log(1+y^{-2}))\log^2(4+|z|)\qquad(z=x+iy,\ y>0).\tag{5.7}$$

For $|y|\ge1$, $\log|F_X(z)|\le\pi|y|+O(\log^2(4+|z|))$ by (5.6), (5.7) and $|F_X(\bar z)|=|F_X(z)|$. For $|y_1|<1$, apply the sub-mean-value property of $\log|F_X|$ on the circle $|z-z_1|=\frac12$. The factor $\log(1+y^{-2})$ is integrable along the circle uniformly in y_1, since it has at most two logarithmic singularities. This gives $\log|F_X(z_1)|\le\frac{3\pi}{2}+O(\log^2(4+|z_1|))$ for $|y_1|<1$. Hence the type is at most π and the indicator satisfies $h(\theta)\le\pi|\sin\theta|$. Since $\log|F_X(iy)|=\pi|y|-O(\log^2(4+|y|))$ for $|y|\ge1$, $h(\pm\frac{\pi}{2})=\pi$. The indicator diagram lies in $[-i\pi,i\pi]$, whose support function is $\pi|\sin\theta|$, and it contains $\pm i\pi$, so it equals that segment. □

5.2. **The criterion.** A weight $w>0$ on $\mathbb{R}$ is in (A_2) if

$$[w]_{A_2}:=\sup_I\Big(\frac{1}{|I|}\int_I w\Big)\Big(\frac{1}{|I|}\int_I w^{-1}\Big)<\infty,\tag{5.8}$$

the supremum being over bounded intervals.

Theorem 5.3 (Block $\frac14$-criterion)**.** *Let X satisfy* (5.1) *and suppose $\psi_X=A+B$ with measurable A,B such that:*

(i) $\|A\|_\infty\le d<\frac14$;
(ii) $\sup_x|\mathcal{Q}_1B(x)-c_B|\le U$ *for some constant c_B.*

Then $w(x):=|F_X(x+i)|^2$ is in (A_2) and

$$[w]_{A_2}\le\frac{\pi^2\coth^2(\pi)\,e^{4\pi U}}{\cos^2(2\pi d)}.\tag{5.9}$$

Proof. $B=\psi_X-A\in L^1(dx/(1+x^2))$, so $\mathcal{Q}_1\psi_X=\mathcal{Q}_1A+\mathcal{Q}_1B$. By (5.6), $w=e^{2c_X''}|F_0(x+i)|^2e^{2\pi\mathcal{Q}_1A}e^{2\pi\mathcal{Q}_1B}$.

Put $\Phi_A(z):=\int A(s)\big[\frac{1}{z-s}+\frac{s}{1+s^2}\big]ds$. It is holomorphic on $\Im z>0$, and $\Phi_A(\zeta+i)$ is holomorphic on $\Im\zeta>-1$. Let $\mathcal{W}(\zeta):=\exp(2\Phi_A(\zeta+i))$. For $\zeta=\xi+i\eta$ with $\eta\ge0$,

$$|\mathcal{W}(\zeta)|=e^{2\pi\mathcal{Q}_{1+\eta}A(\xi)},\qquad\arg\mathcal{W}(\zeta)=-2\pi P_{1+\eta}A(\xi)\in[-2\pi d,2\pi d].\tag{5.10}$$

So $\mathcal{W}$ maps the closed upper half-plane into the sector $|\arg|\le b:=2\pi d<\frac{\pi}{2}$, and $|\mathcal{W}(x)|=e^{2\pi\mathcal{Q}_1A(x)}$ on $\mathbb{R}$.

$\Re\mathcal{W}$ is a positive harmonic function on $\Im\zeta>0$, continuous up to $\mathbb{R}$. By the Herglotz representation it equals $c\eta+\int P_\zeta(t)\,d\nu(t)$ with $c\ge0$, $\nu\ge0$, and $\nu=\Re\mathcal{W}(t)\,dt$ by continuity. Here $P_\zeta(t):=\frac1\pi\frac{\eta}{(t-\xi)^2+\eta^2}$. Hence $\Re\mathcal{W}(\zeta)\ge\int P_\zeta(t)\Re\mathcal{W}(t)\,dt$. For an interval I with centre ξ_I, put $\zeta_I:=\xi_I+i|I|/2$. Then $P_{\zeta_I}(t)\ge(\pi|I|)^{-1}$ for $t\in I$, and

$$\frac{1}{|I|}\int_I|\mathcal{W}|\le\frac{1}{\cos b}\,\frac{1}{|I|}\int_I\Re\mathcal{W}\le\frac{\pi}{\cos b}\Re\mathcal{W}(\zeta_I)\le\frac{\pi}{\cos b}|\mathcal{W}(\zeta_I)|.\tag{5.11}$$

$1/\mathcal{W}$ maps into the same sector, so the same holds for it. Multiplying, $[|\mathcal{W}|]_{A_2}\le\pi^2/\cos^2 b$.

Finally $w = g\,|\mathcal{W}|$ with $g := e^{2c''_X}|F_0(x+i)|^2 e^{2\pi\mathcal{Q}_1 B}$. Since $|\cos\pi(x+i)|^2 = \cos^2(\pi x)\cosh^2\pi + \sin^2(\pi x)\sinh^2\pi \in [\sinh^2\pi, \cosh^2\pi]$ and $e^{2\pi\mathcal{Q}_1 B} \in [e^{2\pi(c_B-U)}, e^{2\pi(c_B+U)}]$, we have $\sup g/\inf g \le \coth^2(\pi) e^{4\pi U}$. For any weight w_1, $[gw_1]_{A_2} \le (\sup g/\inf g)[w_1]_{A_2}$. □

Corollary 5.4 (Riesz bases)**.** *Under the hypotheses of Theorem 5.3:*

(a) *if X is uniformly discrete (in particular without repeated points), $\{e^{ix_k\xi}\}_{k\in\mathbb{Z}}$ is a Riesz basis of $L^2(-\pi,\pi)$;*

(b) *if X is a finite union of uniformly discrete sets, the generalised divided differences of $\{e^{ix_k\xi}\}$ in the sense of* [2] *form a Riesz basis of $L^2(-\pi,\pi)$. Consequently, for every partition of $\mathbb{Z}$ into consecutive blocks $\mathfrak{B}_m$ that are unions of the groups of* [2]*, the subspaces spanned by $\{\xi^j e^{i\lambda\xi}\}$, with $\lambda = x_k$ for $k \in \mathfrak{B}_m$ and j below the multiplicity of λ, form a Riesz basis of subspaces.*

Proof. By Lemma 5.2, F_X is entire of exponential type with indicator diagram of width 2π and zero set X. By Theorem 5.3, $|F_X(x+i)|^2 \in (A_2)$, which is equivalent to the Helson–Szegő condition [6, Ch. VI]. Then (a) is Pavlov's criterion [2, Thm. 1], and the first statement of (b) is [2, Thm. 3]. Both are stated for $L^2(0,2\pi)$, and translation by π multiplies each exponential by a unimodular constant. Grouping the vectors of a Riesz basis into disjoint groups gives a Riesz basis of subspaces. □

5.3. **Kadec, Avdonin, and variable block length.** Write $x_k = k - \frac12 + \delta_k$, $I_k := [k-1, k)$, $\chi_k := \mathbf{1}_{[x_k,\infty)} - \mathbf{1}_{[k-\frac12,\infty)}$, so that $\psi_X = \sum_k \chi_k$ and $\int \chi_k = -\delta_k$.

Proposition 5.5 (Bounded conjugate for block-supported oscillation)**.** *Let $\{J_m\}$ be a partition of $\mathbb{R}$ into consecutive intervals of lengths $N_m \ge 1$. Let $B = \sum_m B_m$ with $\operatorname{supp} B_m \subset J_m$, $\int B_m = 0$ and $|B_m| \le D'_m$, $\sup_m D'_m < \infty$. Put*

$$U_{\mathrm{loc}} := \sup_m \sup_{x\in J_{m-1}\cup J_m\cup J_{m+1}} \Big|\int_{J_m} \frac{(x-s)\,B_m(s)}{\pi((x-s)^2+1)}\,ds\Big|, \qquad L_0 := \sup_{m_0} \sum_{|m-m_0|\ge 2} \frac{N_m^2}{\operatorname{dist}(J_{m_0}, J_m)^2}. \tag{5.12}$$

If $U_{\mathrm{loc}} < \infty$ and $L_0 < \infty$, then $\sup_x |\mathcal{Q}_1 B(x) - c_B| \le 3U_{\mathrm{loc}} + \frac{L_0}{2\pi}\sup_m D'_m$ for a constant c_B.

Proof. The term $\frac1\pi\int \frac{s}{1+s^2} B_m(s)\,ds$ equals $\frac1\pi\int(\frac{s}{1+s^2} - \frac{c_m}{1+c_m^2})B_m(s)\,ds$, where c_m is the centre of J_m. Since $|\frac{d}{ds}\frac{s}{1+s^2}| \le \frac{1}{1+s^2}$, its modulus is at most $D'_m N_m^2/(2\pi(1+\operatorname{dist}(0,J_m)^2))$. If m_0 is the index with $0 \in J_{m_0}$, then $\operatorname{dist}(0,J_m) \ge \operatorname{dist}(J_{m_0}, J_m)$, so these terms are summable by $L_0 < \infty$. Their sum is the constant c_B. Let $x \in J_{m_0}$. The three blocks with $|m-m_0| \le 1$ contribute at most $3U_{\mathrm{loc}}$. For $|m-m_0| \ge 2$, $\int B_m = 0$ gives $\big|\int_{J_m} Q(x-s)B_m(s)\,ds\big| \le D'_m N_m \cdot \frac{N_m}{2}\sup_{s\in J_m}|Q'(x-s)|$ with $Q(u) := \frac{u}{\pi(u^2+1)}$. Moreover $|Q'(u)| \le \frac{1}{\pi u^2}$ and $|x-s| \ge \operatorname{dist}(J_{m_0}, J_m)$. Sum over m. □

Corollary 5.6 (Kadec and Avdonin, with the constant $\frac14$)**.** (a) [7] *If $|\delta_k| \le L < \frac14$ for all k, then $\{e^{ix_k\xi}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*
(b) [1] *If X is uniformly discrete, $\sup_k|\delta_k| \le D$, and for a fixed N the means of δ_k over consecutive blocks of N indices are at most $d < \frac14$ in modulus, then $\{e^{ix_k\xi}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*

Proof. Both statements follow from Theorem 5.7 below: in (a) take $N_m \equiv 1$, and in (b) take $N_m \equiv N$. In both cases $L_0 \le 2\sum_{j\ge1} N^2/(jN)^2 = \pi^2/3$ and $V \le 2D\log(2+3N)$; for $N = 1$, $D_m = 0$ and $V = 0$. For (a), $L < \frac14$ gives gaps $\ge 1 - 2L > \frac12$, so X is uniformly discrete. □

Theorem 5.7 (Avdonin's theorem with variable block length)**.** *Let X be uniformly discrete with $\sup_k|\delta_k| \le D$. Let $\{\mathfrak{B}_m\}$ be a partition of $\mathbb{Z}$ into consecutive blocks of N_m indices, and $J_m := \bigcup_{k\in\mathfrak{B}_m} I_k$. Let $\bar\delta_m$ be the mean of δ_k over $\mathfrak{B}_m$, $D_m := \max_{k\in\mathfrak{B}_m}|\delta_k - \bar\delta_m|$, and $N_m^* := N_{m-1} + N_m + N_{m+1}$. Assume*

(a) $\sup_m|\bar\delta_m| \le d < \frac14$;

(b) $L_0 < \infty$, *with* L_0 *as in* (5.12)*;*
(c) $V := \sup_m D_m \log(2 + N_m^*) < \infty$.

Then $\{e^{ix_k\xi}\}$ *is a Riesz basis of* $L^2(-\pi,\pi)$*, and* (5.9) *holds with*

$$U = 16(1+D)^3 + \frac{6V}{\pi} + \frac{VL_0}{2\pi\log 2}. \tag{5.13}$$

Proof. Write $\psi_X = A + B_1 + B_2$ with $A := -\sum_m \bar\delta_m \mathbf{1}_{J_m}$, $B_1 := \sum_k b_k$ where $b_k := \chi_k + \delta_k \mathbf{1}_{I_k}$, and $B_2 := -\sum_m \sum_{k\in\mathfrak{B}_m} (\delta_k - \bar\delta_m)\mathbf{1}_{I_k}$. Then $\|A\|_\infty \le d$.

B_1. b_k has mean zero, $|b_k| \le 1+D$, and support in an interval of length at most $2D+1$ containing $c_k := k - \frac12$. With Q as above, $\sup|Q| = \frac{1}{2\pi}$ and $|Q'(u)| \le \frac{1}{\pi(1+u^2)}$, so $|\int Q(x-s)b_k(s)\,ds| = |\int (Q(x-s)-Q(x-c_k))b_k(s)\,ds| \le (1+D)(2D+1)\min\bigl(\frac1\pi, (2D+1)\sup_{s\in\operatorname{supp} b_k}|Q'(x-s)|\bigr)$. At most $8D+5 \le 8(1+D)$ indices have $|x-c_k| < 2(2D+1)$, and they contribute at most $16(1+D)^3/\pi$. For the others $|x-s| \ge |x-c_k|/2$ on the support, so the term is at most $4(1+D)^3 \cdot \frac{4}{\pi}(4+(x-c_k)^2)^{-1}$. Since $\sum_k (4+(x-c_k)^2)^{-1} \le \frac12 + \frac{\pi}{2}$, they contribute at most $\frac{16}{\pi}(\frac12+\frac{\pi}{2})(1+D)^3$. The total is below $16(1+D)^3$. The modifying term of $\mathcal{Q}_1$ contributes a convergent sum of constants, as in Proposition 5.5.

B_2. On J_m it has mean zero and modulus at most D_m. For x in $J_{m-1}\cup J_m \cup J_{m+1}$, $|\int_{J_m} Q(x-s)B_2(s)\,ds| \le \frac{D_m}{\pi}\int_{|u|\le N_m^*} \frac{|u|\,du}{1+u^2} \le \frac{2}{\pi} D_m \log(2+N_m^*) \le \frac{2V}{\pi}$. Moreover $\sup_m D_m \le V/\log 2$. Proposition 5.5 gives $|\mathcal{Q}_1 B_2 - c| \le \frac{6V}{\pi} + \frac{VL_0}{2\pi\log 2}$.

Apply Theorem 5.3 and Corollary 5.4(a). □

Remark 5.8 (The price of variable length). For fixed N, conditions (b) and (c) hold automatically, and Theorem 5.7 is Avdonin's theorem. If $N_m \to \infty$, the averaging condition (a) is still only $|\bar\delta_m| \le d < \frac14$. However, (c) then requires the oscillation inside the block to decay like $1/\log N_m$. In Theorem 5.3 the oscillation enters only through the bounded-conjugate condition (ii), which allows cancellation inside the blocks and is weaker than (c). Without some condition of this kind the conclusion fails for growing blocks: a block of length N with oscillation of size D_m can make $|\mathcal{Q}_1 B|$ of size $D_m \log N$. Lyubarskii and Seip derive Kadec's theorem from their Muckenhoupt characterisation ([8, Thm. 2] for $p = 2$), and indicate that the same technique gives results of Avdonin type. I have not searched the literature for Theorem 5.7 or for the bounded-conjugate formulation of Theorem 5.3, so their novelty is not claimed. The proof of Theorem 5.3 is the sufficiency half of the Helson–Szegő argument, applied on the line $\Im z = 1$.

6. Application to lists near the RvM clock

6.1. **Unfolding.**

Definition 6.1 (Unfolded list)**.** Let $t_0 := 9.6769\ldots$ be the zero of $\mathcal{N}_{\mathrm{RVM}}$ in $(2\pi,\infty)$. For a C-admissible list $\boldsymbol{\kappa}$ with $\kappa_1 > t_0$ (true for $\boldsymbol{\tau}$ and $\boldsymbol{\gamma}$), put $x_k := \mathcal{N}_{\mathrm{RVM}}(\kappa_k) > 0$ for $k \ge 1$ and $x_{1-k} := -x_k$. The resulting symmetric sequence $X_{\boldsymbol{\kappa}}$ has discrepancy $\psi_{\boldsymbol{\kappa}} := \psi_{X_{\boldsymbol{\kappa}}}$, and $F_{\boldsymbol{\kappa}} := F_{X_{\boldsymbol{\kappa}}}$ with $z_0 = 0$. For the clock, $X_{\boldsymbol{\tau}} = \mathbb{Z} + \frac12$ and $F_{\boldsymbol{\tau}} = F_0$.

Lemma 6.2 (Unfolded discrepancy)**.** *For a* C*-admissible list with* $\kappa_1 > t_0$*,* $|x_k - (k-\frac12)| \le \frac{C}{2\pi}\log\frac{\tau_k + C}{2\pi}$ *for* $k \ge 1$*, so* (5.1) *holds. Moreover, for* $x = \mathcal{N}_{\mathrm{RVM}}(t)$ *with* $x > 0$*,*

$$\psi_{\boldsymbol{\kappa}}(x) = N_{\boldsymbol{\kappa}}(t) - N_\tau(t), \qquad |N_{\boldsymbol{\kappa}}(t) - N_\tau(t)| \le \frac{C}{\pi}\log\frac{t+C}{2\pi} + 2, \tag{6.1}$$

and $\psi_{\boldsymbol{\kappa}}(-x) = -\psi_{\boldsymbol{\kappa}}(x)$ *for almost every* x*.*

Proof. By the mean value theorem and the monotonicity of $\mathcal{N}'_{\mathrm{RVM}}$, $|\mathcal{N}_{\mathrm{RVM}}(\kappa_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)| \le C\mathcal{N}'_{\mathrm{RVM}}(\tau_k + C)$. For $x > 0$ the terms of (5.2) with $k \le 0$ vanish, because both x_k and $k - \frac12$ are negative. For $k \ge 1$, $x_k \le x$ if and only if $\kappa_k \le t$, by monotonicity of $\mathcal{N}_{\mathrm{RVM}}$. If $N_{\boldsymbol{\kappa}}(t) = j$, then

$\tau_j \le t+C$ and $\tau_{j+1} > t-C$, so $N_\tau(t-C) \le j \le N_\tau(t+C)$. By (2.2), $N_\tau(t+C)-N_\tau(\max(t-C,2\pi)) \le \mathcal{N}_{\mathrm{RVM}}(t+C) - \mathcal{N}_{\mathrm{RVM}}(\max(t-C,2\pi)) + 1$. This is at most $\frac{2C}{2\pi}\log\frac{t+C}{2\pi} + 2$, the extra 1 covering $t - C < 2\pi$, where $\mathcal{N}_{\mathrm{RVM}} \ge -\frac{1}{8}$ and $N_\tau = 0$. Oddness follows from the symmetry of $X_{\boldsymbol{\kappa}}$ and of $\mathbb{Z} + \frac{1}{2}$. □

Proposition 6.3 (The condition of $\mathscr{O}$ gives the averaging hypothesis with blocks $\asymp \log^2 T$)**.** *Let $\boldsymbol{\kappa}$ be C-admissible with $\kappa_1 > t_0$, and assume the third condition of $\mathscr{O}$ (`eq: list-space`) in the form*

$$\Big|\int_{2\pi}^{X} (N_{\boldsymbol{\kappa}}(t) - \mathcal{N}_{\mathrm{RVM}}(t))\,dt\Big| \le C_2 \log X \qquad (X \ge 3). \tag{6.2}$$

Put $H(T) := 16C_2 \log T$. There is $T_1 = T_1(C, C_2)$ such that the following holds for the points $T_{m+1} := T_m + H(T_m)$ ($m \ge 1$) and the blocks $J_m := [\mathcal{N}_{\mathrm{RVM}}(T_m), \mathcal{N}_{\mathrm{RVM}}(T_{m+1}))$ with their mirror images $-J_m$. The step function $A_{\boldsymbol{\kappa}}$, equal to the mean of $\psi_{\boldsymbol{\kappa}}$ on each $\pm J_m$ and to 0 on $(-\mathcal{N}_{\mathrm{RVM}}(T_1), \mathcal{N}_{\mathrm{RVM}}(T_1))$, satisfies

$$\|A_{\boldsymbol{\kappa}}\|_\infty \le \tfrac{3}{16}, \qquad |J_m| = \frac{8C_2}{\pi}\log^2 T_m\,(1 + o(1)) \quad (m \to \infty). \tag{6.3}$$

The remainder $B_{\boldsymbol{\kappa}} := \psi_{\boldsymbol{\kappa}} - A_{\boldsymbol{\kappa}}$ has mean zero on every block. For $\boldsymbol{\kappa} = \boldsymbol{\gamma}$, (6.2) *holds unconditionally for some C_2.*

Proof. Write $\mathcal{N} = \mathcal{N}_{\mathrm{RVM}}$ and $J = [\mathcal{N}(T), \mathcal{N}(T+H))$. Then

$$\begin{aligned}\int_J \psi_{\boldsymbol{\kappa}}\,dx &= \int_T^{T+H} (N_{\boldsymbol{\kappa}} - N_\tau)(t)\,\mathcal{N}'(t)\,dt \\ &= \mathcal{N}'(T)\int_T^{T+H} (N_{\boldsymbol{\kappa}} - N_\tau) + \int_T^{T+H} (N_{\boldsymbol{\kappa}} - N_\tau)(\mathcal{N}'(t) - \mathcal{N}'(T))\,dt.\end{aligned} \tag{6.4}$$

The last integral is at most $H \cdot \frac{H}{2\pi T}\big(\frac{C}{\pi}\log\frac{T+H+C}{2\pi} + 1\big)$, by (6.1) and $\mathcal{N}''(t) = \frac{1}{2\pi t}$.

Next, $\int_T^{T+H}(N_{\boldsymbol{\kappa}} - N_\tau) = \int_T^{T+H}(N_{\boldsymbol{\kappa}} - \mathcal{N}) + \int_T^{T+H}(\mathcal{N} - N_\tau)$. The first piece is at most $2C_2\log(T+H)$ by (6.2). For the second, substitute $x = \mathcal{N}(t)$. By (2.2), $\mathcal{N} - N_\tau = \mathrm{saw}(x) := x - \lfloor x + \frac{1}{2}\rfloor$, which is 1-periodic with mean zero and $|\mathrm{saw}| \le \frac{1}{2}$, and $dt = g(x)\,dx$ with $g := 1/\mathcal{N}'(t(x))$ positive and decreasing. On a full period $[j-\frac{1}{2}, j+\frac{1}{2})$, $|\int \mathrm{saw}\, g| = |\int \mathrm{saw}\,(g - g(j))| \le \frac{1}{4}(g(j-\frac{1}{2}) - g(j+\frac{1}{2}))$, which telescopes. The two partial periods contribute at most $g(\mathcal{N}(T))$. Hence $|\int_T^{T+H}(\mathcal{N} - N_\tau)| \le \frac{5}{4\mathcal{N}'(T)}$.

Since $|J| \ge \mathcal{N}'(T)H$,

$$\Big|\frac{1}{|J|}\int_J \psi_{\boldsymbol{\kappa}}\Big| \le \frac{2C_2\log(T+H)}{H} + \frac{5}{4\mathcal{N}'(T)H} + \frac{H}{2\pi T\mathcal{N}'(T)}\Big(\frac{C}{\pi}\log\frac{T+H+C}{2\pi} + 1\Big). \tag{6.5}$$

With $H = 16C_2\log T$, the first term tends to $\frac{1}{8}$ and the others tend to 0, so the bound is at most $\frac{3}{16}$ for $T \ge T_1(C, C_2)$. The length is $|J_m| = \mathcal{N}'(\xi)H(T_m)$ with $\xi \in [T_m, T_{m+1}]$. The mirror blocks follow by oddness. For $\boldsymbol{\gamma}$: $N_{\boldsymbol{\gamma}}(t) = 1 + \vartheta(t)/\pi + S(t)$ [11, §9.3], $\int_0^X S = O(\log X)$ [11, Thm. 9.9], and $1 + \vartheta/\pi - \mathcal{N}_{\mathrm{RVM}} = O(1/t)$. Finally, $\boldsymbol{\gamma}$ is C-admissible for some C, unconditionally. By [11, §9.3, Thm. 9.4], $|N_{\boldsymbol{\gamma}} - N_\tau| \le A\log t$ for $t \ge t_1$, with constants A and t_1. If $\gamma_k \ge 2\tau_k$ with $\tau_k \ge t_1$, then $N_\tau(2\tau_k) - N_{\boldsymbol{\gamma}}(2\tau_k) \ge \mathcal{N}_{\mathrm{RVM}}(2\tau_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k) - 1 \gg \tau_k\log\tau_k$, which is impossible for large k. Otherwise the argument in the proof of Corollary 4.9, with $\Phi(T) = A\log T$ and $d \le \tau_k$, gives $|\gamma_k - \tau_k| \le 2\pi A\log(2\tau_k)/\log((\tau_k - d)/2\pi)$, which is bounded for large k. The finitely many remaining k give a finite supremum. In particular (5.1) holds for $X_{\boldsymbol{\gamma}}$, by Lemma 6.2. □

Remark 6.4 (Correction of an earlier formulation)*.* Proposition 6.3 is an implication. The averaging condition on blocks of length $\asymp \log^2 T$ does not imply (6.2): summing $|\int_{J_m}\psi| \le \frac{3}{16}|J_m|$ over the blocks up to height X gives only $O(X\log X)$, not $O(\log X)$. The two conditions do not coincide.

Corollary 6.5 (Reduction to one condition)**.** *For every $\boldsymbol{\kappa}$ as in Proposition 6.3: if $\mathcal{Q}_1 B_{\boldsymbol{\kappa}}$ is bounded up to a constant, then $|F_{\boldsymbol{\kappa}}(x+i)|^2 \in (A_2)$, with constant* (5.9) *for $d = \frac{3}{16}$. Corollary 5.4(a) then holds for $X_{\boldsymbol{\kappa}}$ if it is uniformly discrete, and Corollary 5.4(b) if it is a finite union of uniformly discrete sets.*

Proof. Lemma 6.2, Proposition 6.3 and Theorem 5.3. □

6.2. **The weight is a quotient of synthetic zetas.**

Proposition 6.6 (Weight identity)**.** *Let $\boldsymbol{\kappa}$ be C-admissible and satisfy* (6.2)*. For $0 < h \le 1$ and $t \ge 2\pi$,*

$$\log\Big|\frac{\zeta_{\boldsymbol{\kappa}}}{\zeta_{\mathrm{RVM}}}\Big(\tfrac{1}{2}+h+it\Big)\Big| = \int_{2\pi}^{\infty} \frac{(t-s)\,(N_{\boldsymbol{\kappa}}-N_\tau)(s)}{(t-s)^2+h^2}\,ds + c_{\boldsymbol{\kappa}} + m_{\boldsymbol{\kappa},h}(t), \tag{6.6}$$

where the integral converges as an improper integral and $c_{\boldsymbol{\kappa}}$ is a constant. The remainder satisfies $|m_{\boldsymbol{\kappa},h}(t)| \le c(C,C_2)\log(2+t)/(1+t)$.

Proof. For $s = \frac{1}{2}+h+it$, $|\frac{1}{4}+\kappa^2+q(s)| = |\kappa^2+(h+it)^2| = |\kappa - t + ih|\,|\kappa + t - ih|$. Hence, with $\ell_h(u) := \frac{1}{2}\log(u^2+h^2)$,

$$\log\Big|\frac{P_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\tau}}}(s)\Big| = \sum_k \big[\ell_h(t-\kappa_k) - \ell_h(t-\tau_k)\big] + \sum_k \big[\ell_h(t+\kappa_k) - \ell_h(t+\tau_k)\big] - \sum_k \log\frac{\frac{1}{4}+\kappa_k^2}{\frac{1}{4}+\tau_k^2}. \tag{6.7}$$

Put $g := N_{\boldsymbol{\kappa}} - N_\tau$ and $\mathcal{G}(s) := \int_{2\pi}^{s} g$. By (6.2), by the sawtooth estimate in the proof of Proposition 6.3 on $[\tau_1,\infty)$, and by $|\mathcal{N}_{\mathrm{RVM}} - N_\tau| \le \frac{1}{2}$ on $[2\pi,\tau_1]$, $|\mathcal{G}(s)| \le c\log(2+s)$. For f smooth with $f'(s) = O(1/s)$ and $f''(s) = O(1/s^2)$, the sums $\sum_{k\le K}[f(\kappa_k) - f(\tau_k)] = -\int f' g_K$ converge as $K\to\infty$ to $-\int^\infty f'g$, an improper integral. Integration by parts against $\mathcal{G}$ shows both the convergence and that the tail contributions vanish. With $f = \ell_h(t-\cdot)$ this gives the integral in (6.6). With $f = \log(\frac{1}{4}+(\cdot)^2)$ it gives the constant $c_{\boldsymbol{\kappa}}$. With $f = \ell_h(t+\cdot)$ it gives $m_{\boldsymbol{\kappa},h}(t) = \int_{2\pi}^{\infty} k'(s)\mathcal{G}(s)\,ds$, where $k(s) := \frac{t+s}{(t+s)^2+h^2}$ and $|k'(s)| \le (t+s)^{-2}$. This yields the bound. □

Remark 6.7 (Local scale). With $h = 1$ and $h_T := h/\mathcal{N}'_{\mathrm{RVM}}(T) = 2\pi/\log(T/2\pi)$ (the notation of the proof of Theorem 6.8), the substitution $x = \mathcal{N}_{\mathrm{RVM}}(t)$ maps the kernel $\frac{(t-s)\,ds}{(t-s)^2+h_T^2}$ to $\frac{(x-x')\,dx'}{(x-x')^2+1}$ to first order near $t = T$. Comparing (6.6) with (5.6) at $y = 1$, the unfolded weight $|F_{\boldsymbol{\kappa}}(x+i)|^2$ near $x = \mathcal{N}_{\mathrm{RVM}}(T)$ is therefore, to first order, $|\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}|^2$ on the line $\Re s = \frac{1}{2}+h_T$, up to the bounded factor $|F_0(x+i)|^2$. The identity (6.6) is exact. The identification with the unfolded weight is local, and its quantitative form for the Poisson part is Step 3 of the proof of Theorem 6.8.

6.3. **Obstruction for the zeta list.**

Theorem 6.8 (No (A_2) for the unfolded zeta ordinates)**.** *Let $X_{\boldsymbol{\gamma}}$ be the unfolded list of the ordinates $\gamma_k > 0$ of the nontrivial zeros of ζ, counted with multiplicity, and let F be any entire function of exponential type whose zero set is $X_{\boldsymbol{\gamma}}$, with multiplicities. Then $|F(x+ih)|^2 \notin (A_2)$ for every real $h \ne 0$. The proof does not assume the Riemann hypothesis.*

Proof. Write $\psi := \psi_{\boldsymbol{\gamma}}$, $\mathcal{N} := \mathcal{N}_{\mathrm{RVM}}$, and $F_{\boldsymbol{\gamma}}$ for the function of Definition 6.1. It is even and real by Lemma 5.2(iii), because $x_1 = \mathcal{N}(\gamma_1) = 0.45\ldots > 0$, so $0 \notin X_{\boldsymbol{\gamma}}$.

Step 1: reduction to $F_{\boldsymbol{\gamma}}$ and $h > 0$. $F/F_{\boldsymbol{\gamma}}$ is entire, zero-free and of exponential type, so it equals e^{a+bz} by Hadamard's theorem. Hence $|F(x+ih)|^2 = c\,e^{2\beta x}w_0(x)$ with $\beta = \Re b$, a constant $c > 0$, and $w_0(x) := |F_{\boldsymbol{\gamma}}(x+ih)|^2$, which is even in x. Moreover $|F_{\boldsymbol{\gamma}}(x-ih)| = |F_{\boldsymbol{\gamma}}(x+ih)|$. Suppose $e^{2\beta x}w_0 \in (A_2)$. Reflection preserves (A_2), so $e^{-2\beta x}w_0 \in (A_2)$ as well. By Step 2 below both logarithms are in BMO, so $4\beta x \in \mathrm{BMO}$, which forces $\beta = 0$. So we may assume $F = F_{\boldsymbol{\gamma}}$ and $h > 0$, and that $w_0 \in (A_2)$.

Step 2: BMO. Let $f := \log w_0$ and f_I its mean on I. By Jensen's inequality, $\frac{1}{|I|}\int_I e^{\pm(f-f_I)} \geq 1$. The product of the two averages is at most $[w_0]_{A_2}$, so each is at most $[w_0]_{A_2}$. Since $u^2 \leq 2e^{|u|} \leq 2(e^u + e^{-u})$,

$$\frac{1}{|I|}\int_I |f - f_I|^2 \leq 4[w_0]_{A_2} \qquad \text{for every interval } I. \tag{6.8}$$

By (5.6) and the boundedness of $\log|F_0(x+ih)|$, $u := \pi\mathcal{Q}_h\psi \in \mathrm{BMO}$. Now take $\Phi_h(z) := \Phi_X(z+ih)$, in the notation of the proof of Lemma 5.2, which is holomorphic on $\Im z > -h$. Its real part on $\mathbb{R}$ is u, and its imaginary part on $\mathbb{R}$ is $-\pi P_h\psi$. By (5.7), $\Re\Phi_h(z) = O(\log^2(4+|z|))$ on $\Im z \geq 0$, and so is the Poisson integral $P[u]$. Hence $\Re\Phi_h - P[u]$ is harmonic on $\Im z > 0$, continuous with zero boundary values, and $O(\log^2|z|)$. After odd reflection it is harmonic on $\mathbb{C}$ and $o(|z|)$, hence constant, hence zero. Therefore $\Phi_h = P[u] + i\mathcal{Q}[u] + ic$ on $\Im z > 0$, where $P[u] + i\mathcal{Q}[u]$ is the holomorphic function with real part $P[u]$ and $\mathcal{Q}[u](x+iy) = \mathcal{Q}_y u(x)$. Letting $y \to 0$, $-\pi P_h\psi = \tilde{u} + c$ almost everywhere on $\mathbb{R}$, where $\tilde{u}$ is the conjugate function of u, the a.e. boundary value of $\mathcal{Q}_y u$ [6, Ch. III]. Conjugation is bounded on BMO [6, Ch. VI], so $P_h\psi \in \mathrm{BMO}$. By the John–Nirenberg inequality [6, Ch. VI], there is $K < \infty$ with

$$\frac{1}{|I|}\int_I \bigl|P_h\psi - (P_h\psi)_I\bigr|^2 \leq K \qquad \text{for every interval } I. \tag{6.9}$$

Step 3: back to the ordinates. By [11, §9.3], $N_\gamma(t) = 1 + \vartheta(t)/\pi + S(t)$, with $S(t) = O(\log t)$ [11, Thm. 9.4]. Lemma 6.2 and (2.2) then give, for $x = \mathcal{N}(t) > 0$, $\psi(x) = S(t) + e(x)$ with $|e| \leq c_e$. Here c_e bounds $|1 + \vartheta/\pi - \mathcal{N}| + \frac{1}{2}$ on $t \geq 2\pi$. Let $t_0 := \mathcal{N}^{-1}(0)$ and $T \to \infty$, with $t \in [T, 2T]$ and $h_t := h/\mathcal{N}'(t)$, $h_T := h/\mathcal{N}'(T)$.

(a) The part of $P_h\psi(\mathcal{N}(t))$ coming from $x' < 0$ is $O(\log T/T)$, since $|\psi(x')| \leq c\log(2 + |x'|)$. The part coming from e is at most c_e. Substituting $x' = \mathcal{N}(s)$, the rest is $\int_{t_0}^\infty K_t(s)S(s)\,ds$ with $K_t(s) := P_h(\mathcal{N}(t) - \mathcal{N}(s))\mathcal{N}'(s)$, where $P_h(v) := \frac{h}{\pi(v^2+h^2)}$.

(b) Put $k_t(s) := P_{h_t}(t-s) = P_h(\mathcal{N}'(t)(t-s))\mathcal{N}'(t)$. For $|s-t| \leq T^{1/3}$, Taylor's formula with $\mathcal{N}'' = \frac{1}{2\pi s}$ gives $|\mathcal{N}(t) - \mathcal{N}(s) - \mathcal{N}'(t)(t-s)| \leq \delta_T := T^{-1/3}/\pi$. Hence $|K_t(s) - k_t(s)| \leq (5\delta_T/h)k_t(s)$ for T large. With $|S| \leq c\log(3T)$ on that range, the core difference is $O(T^{-1/3}\log T)$. For $|s-t| > T^{1/3}$, both kernels carry mass $O(T^{-1/3})$ against a weight $O(\log(2+s))$, because $|\mathcal{N}(t) - \mathcal{N}(s)| \geq T^{1/3}$ there when $\mathcal{N}' \geq 1$. So that part tends to 0 as well.

(c) $\|P_a - P_b\|_{L^1} \leq \frac{2}{\pi}|\log(a/b)|$, because $\int|\partial_c P_c| = \frac{2}{\pi c}$. Since $h_t/h_T \in [1 - \frac{2\log 2}{\log(T/2\pi)}, 1]$, $|\int(P_{h_t} - P_{h_T})(t-s)S(s)\,ds| = O(\log T/\log T) = O(1)$, again with the tails controlled by $|S(s)| \leq c\log(2+s)$.

(d) $\|P_{h_T} * S - S\|_{L^2[T,2T]} = O(\sqrt{T})$, with $S := 0$ on $(-\infty, t_0)$; this is Lemma 6.9 below.

Together, (a)–(d) give $\|P_h\psi \circ \mathcal{N} - S\|_{L^2[T,2T]} = O(\sqrt{T})$.

Step 4: contradiction. Let $I_T := [\mathcal{N}(T), \mathcal{N}(2T)]$ and $c_T := (P_h\psi)_{I_T}$. By (6.9), $\int_T^{2T}|P_h\psi(\mathcal{N}(t)) - c_T|^2\mathcal{N}'(t)\,dt \leq K|I_T| \leq KT\mathcal{N}'(2T)$, so $\int_T^{2T}|P_h\psi(\mathcal{N}(t)) - c_T|^2\,dt \leq 2KT$. With Step 3, $\int_T^{2T}|S - c_T|^2 = O(T)$. But

$$\int_T^{2T}|S-c|^2 \geq \int_T^{2T} S^2 - \frac{1}{T}\Bigl(\int_T^{2T} S\Bigr)^2 = \frac{1}{2\pi^2}T\log\log T\,(1+o(1)) \tag{6.10}$$

for every constant c, by Selberg's theorem and $\int_T^{2T} S = O(\log T)$ [11, Thm. 9.9]. This is a contradiction. □

Lemma 6.9 (Smoothing S at scale $1/\log T$)**.** *Let $h > 0$ be fixed, $h_T := 2\pi h/\log(T/2\pi)$, and $S := 0$ on $(-\infty, t_0)$. Then $\|P_{h_T} * S - S\|_{L^2[T,2T]} = O_h(\sqrt{T})$, unconditionally.*

Proof. By Fujii's Main Theorem [3, Main Theorem], proved unconditionally by modifying Selberg's method, applied with $q=1$, $H=T$, $k=1$: for $0<u\le 1$ and $T\ge T_0$,

$$\int_T^{2T}\bigl(S(t+u)-S(t)\bigr)^2dt=\frac{T}{\pi^2}\log(2+u\log T)\Bigl(1+O\bigl((\log(2+u\log T))^{-1/2}\bigr)\Bigr)\le c_F T\log(2+u\log T). \tag{6.11}$$

The range $0<u\le H-(H/\sqrt{T})^{1/8}=T-T^{1/16}$ of [3] contains $0<u\le 1$, and the same bound holds for $-1\le u<0$: the substitution $t\mapsto t-u$ moves the integral to $[T-|u|,2T-|u|]\subset[T/2,2T]$, covered by (6.11) at the base points $T/2$ (with $H=T/2$) and T (with $H=T$). By Minkowski's inequality, $\|P_{h_T}*S-S\|_{L^2[T,2T]}\le\int P_{h_T}(u)\|S(\cdot-u)-S\|_{L^2[T,2T]}\,du$. For $0<|u|\le 1$ the norm is at most $(2c_FT\log(2+|u|\log T))^{1/2}$. For $1<|u|\le T/2$, Selberg's theorem $\int_0^T S^2=\frac{1}{2\pi^2}T\log\log T+O(T(\log\log T)^{1/2})$ [10] bounds it by $c(T\log\log T)^{1/2}$. For $|u|>T/2$, $|S(t)|\le c\log(2+t)$ [11, Thm. 9.4] bounds it by $c\sqrt{T}\log(3T+|u|)$. Substituting $u=h_Tv$ in the first range, where $h_T\log T\to 2\pi h$, and using $\int_{|u|>1}P_{h_T}=O(h_T)$ and $\int_{|u|>T/2}P_{h_T}(u)\log(3T+|u|)\,du=O(h_T\log T/T)$, all three contributions are $O(\sqrt{T})$. □

Remark 6.10 (A second proof). Lemma 6.9 also follows from Selberg's mean-square approximation of $S(t)$ by $S_x(t):=-\frac{1}{\pi}\sum_{p\le x}p^{-1/2}\sin(t\log p)$, $x=T^\delta$ [10], and the Montgomery–Vaughan mean value theorem [9, Cor. 3]. Indeed $P_{h_T}*S_x-S_x=-\frac{1}{\pi}\sum_{p\le x}p^{-1/2}(e^{-h_T\log p}-1)\sin(t\log p)$ has mean square $\ll T\,h_T^2\log^2 x\ll T$ on $[T,2T]$, and the remainder $S-S_x$ is handled by Minkowski's inequality. Fujii's announcement [4] states the moment lemma of [3] in the form used in [5, Lemma 1].

Corollary 6.11 (Consequences for the zeta list)**.** (a) *$\{e^{ix_k\xi}\}_{x_k\in X_\gamma}$ is not a Riesz basis of $L^2(-\pi,\pi)$.*
(b) *If X_γ is a finite union of uniformly discrete sets, its generalised divided differences are not a Riesz basis of $L^2(-\pi,\pi)$.*
(c) *For every decomposition $\psi_\gamma=A+B$ with $\|A\|_\infty<\frac14$, $\mathcal{Q}_1B$ is unbounded. In particular this holds for the blocks of Proposition 6.3, where the averaging hypothesis is satisfied.*

Proof. (a) and (b): the criteria [2, Thm. 1, Thm. 3] are necessary and sufficient, and require a generating function satisfying Helson–Szegő, equivalently (A_2), on some line $\Im z=h$. The line $h=0$ is excluded, since $|F(x)|^{-2}$ is not locally integrable at a real zero. All lines with $h\neq 0$ are excluded by Theorem 6.8. (c): otherwise Theorem 5.3 would give $|F_\gamma(x+i)|^2\in(A_2)$. □

Heuristic 6.12 (Graduated version)**.** The proof of Theorem 6.8 uses intervals I_T of length $\asymp T\log T$. On a window of L unfolded points at height T, the mean square oscillation of $P_h\psi$ is governed by the variance of increments of S over t-intervals of length $L/\mathcal{N}'(T)$. By Fujii's asymptotics this is expected to be of size $\frac{1}{\pi^2}\log L$. This suggests that local (A_2) constants on windows of L points grow with L but stay bounded for L bounded. That is the graduated stability that a Fredholm formulation would need. None of this is proved here.

Remark 6.13 (Bearing on route B). Theorem 6.8 rules out a uniform Riesz-basis (Fredholm) structure for the unfolded zeta list on the whole line, and hence a linear stability argument with constants independent of the height. It says nothing about statement B. The compactness step of route B (Proposition 3.4) uses no stability, and the open part of route B is existence (Definition 3.1) together with uniqueness in the box (Hypothesis 3.5).

7. Status of the statements

Statement	Status
Lemma 3.2, Corollary 3.3	proved; conditional on (D) as stated
Proposition 3.4	proved, unconditional

Hypothesis 3.5	open; it extends `thm:free-masses`; not needed with prescribed masses (Proposition 3.10)
Corollary 3.7	proved, assuming B and U
Hypothesis 3.9	to be checked against `thm:distributional-EF`
Proposition 3.10, Corollary 3.12	proved, assuming Hypothesis 3.9
Hypothesis 3.15	open; implies the Riemann hypothesis
Proposition 3.16	proved
Lemmas 4.2–4.7, Theorem 4.8, Corollary 4.9	proved, unconditional; uses `thm:halfplane-RVM`
Heuristics 4.11, 4.12, 6.12	not proved
Lemma 5.2, Theorem 5.3, Corollary 5.4	proved; Corollary 5.4 uses [2]
Corollary 5.6, Theorem 5.7, Proposition 5.5	proved; novelty not checked
Lemma 6.2, Proposition 6.3, Proposition 6.6, Corollary 6.5	proved, unconditional
Lemma 6.9, Theorem 6.8, Corollary 6.11	proved, unconditional; uses [3, 10, 11, 6]

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Email address: `rotkivanyasac@gmail.com`